

Foundation (Jalapeno)

Q1. Rationalize the denominator of $\frac{1}{\sqrt{2}}$.

- (A) $\frac{\sqrt{2}}{2}$
- (B) $\frac{1}{2\sqrt{2}}$
- (C) $\frac{2}{\sqrt{2}}$
- (D) $\frac{1}{2}$
- (E) $\frac{\sqrt{2}}{4}$

Q2. A stock's price is $\frac{1}{\sqrt{3}}$ times its projected price. If the projected price is \$45, what is the current price?

- (A) $15\sqrt{3}$
- (B) \$15
- (C) $45\sqrt{3}$
- (D) $\frac{45}{\sqrt{3}}$
- (E) $\frac{45\sqrt{3}}{3}$

Q3. Rationalize the denominator of $\frac{2}{\sqrt{5}-1}$.

- (A) $\frac{2(\sqrt{5}+1)}{4}$
- (B) $\frac{2(\sqrt{5}-1)}{4}$
- (C) $\frac{2(\sqrt{5}+1)}{5-1}$
- (D) $\frac{2(\sqrt{5}-1)}{5-1}$
- (E) $\frac{2(\sqrt{5}+1)}{5+1}$

Q4. A business's profit is $\frac{3}{\sqrt{2}}$ times its investment. If the investment is \$1000, what is the profit?

- (A) $1500\sqrt{2}$
- (B) \$1500
- (C) $2100\sqrt{2}$
- (D) $\frac{3000}{\sqrt{2}}$
- (E) $\frac{3000\sqrt{2}}{2}$

Q5. Rationalize the denominator of $\frac{5}{2+\sqrt{3}}$.

- (A) $\frac{5(2-\sqrt{3})}{1}$
- (B) $\frac{5(2+\sqrt{3})}{1}$
- (C) $\frac{5(2-\sqrt{3})}{4-3}$
- (D) $\frac{5(2+\sqrt{3})}{4-3}$
- (E) $\frac{5(2-\sqrt{3})}{4+3}$

Q6. The value of a stock is $\frac{4}{\sqrt{6}}$ times its price. If the stock's price is \$80, what is its value?

- (A) $160\sqrt{6}$
- (B) \$160
- (C) $320\sqrt{6}$
- (D) $\frac{320}{\sqrt{6}}$
- (E) $\frac{320\sqrt{6}}{6}$

Q7. Rationalize the denominator of $\frac{3}{1-\sqrt{2}}$.

- (A) $\frac{3(1+\sqrt{2})}{1-2}$
- (B) $\frac{3(1-\sqrt{2})}{1-2}$
- (C) $\frac{3(1+\sqrt{2})}{1+2}$
- (D) $\frac{3(1-\sqrt{2})}{1+2}$
- (E) $\frac{3(1+\sqrt{2})}{-1}$

Q8. A company's revenue is $\frac{2}{\sqrt{5}}$ times its expenditure. If the expenditure is \$500, what is the revenue?

- (A) $200\sqrt{5}$
- (B) \$200
- (C) $1000\sqrt{5}$
- (D) $\frac{1000}{\sqrt{5}}$
- (E) $\frac{1000\sqrt{5}}{5}$

Open Ended Questions (FRQ)

Q9. Rationalize the denominator of $\frac{\sqrt{7}+2}{\sqrt{3}-1}$ and simplify.

Q10. A mutual fund's return is $\frac{5}{\sqrt{2}+1}$ times its investment. If the investment is \$2500, find the return and rationalize the denominator.

Chef's Tips

- Check for common factors before rationalizing
- Use the conjugate to rationalize the denominator
- Simplify the expression after rationalizing the denominator